UNIT-5: Data Visualization using Dataframe

5.1 Data Visualization

- Data visualization is representation of data in graphical form.
- It summarizes the data and helps to give outcomes in graphical form.
- The graphical data representation helps people to understand data easily.
- Data can be represented in form of LINE graph, BAR graph, PIE graph, SCATTER graphs etc.
- Python has separate library (package) named MATPLOTLIB which allows data visualization.

5.2 MATPLOTLIB

- Matplotlib is a library for creating static, animated, and interactive visualizations in Python.
- It is a library for data visualization in form of plots, graphs and charts.
- It can plot data directly from Pandas DataFrame.
- Matplotlib library haspyplotmodule which is a collection of functions using which the graph can be plotted in form of bar, line, scatter graphs etc.
- Using these functions, the plotted graph can also be decorated by providing different parameters.
- There is a need to import matplotlib.pyplot into python file for plotting graphs.

5.3 Plotting graph: (only two dimensional Plots)

5.3.1 plot() function

- To plot graph, plot() is used.
- plot() is used to draw points in diagram. By default, it draws lines between these plotted points.
- Syntax:

  plot(x_values, y_values, optional_parameter)
- Optional parameters:
  - color='color_code' : To set color.

character	color
'b'	blue
'g'	green
'r'	red
'c'	cyan
'm'	magenta
'y'	yellow
'k'	black
'w'	white

- marker='point_symbol' : To change plotted point symbol.

'.'	point marker
','	pixel marker
'o'	circle marker
'v'	triangle_down marker
'^'	triangle_up marker
'<'	triangle_left marker
'>'	triangle_right marker
'1'	tri_down marker
'2'	tri_up marker
'3'	tri_left marker
'4'	tri_right marker
's'	square marker
'p'	pentagon marker
'*'	star marker
'h'	hexagon1 marker
'H'	hexagon2 marker
'+'	plus marker
'x'	x marker
'D'	diamond marker
'd'	thin_diamond marker

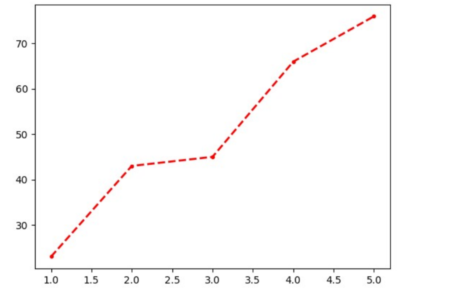
- linestyle='symbol' : To change line style.

character	description
'-'	solid line style
'--'	dashed line style
'-.'	dash-dot line style
':'	dotted line style

- [marker][Linestyle][color]: To set plot symbol, line style and color with one text. Ex: '.b--'
- linewidth=no : To change line width.
- label='text' : To set label which is display with LEGEND()

- Example:

```
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[23,43,45,66,76]
plt.plot(x,y,'.r--,linewidth=2,label='Testing')
plt.show()
```



5.3.2 show() function

- show() function starts an event loop, looks for all currently active figure objects, and opens one or more interactive windows that display your figure or figures.
- Syntax: plt_obj.show()

5.3.3 xlabel() functions

- xlabel() is used to set the label for x-axis.
- Syntax: plt_obj.xlabel('label')

5.3.4 ylabel() function

- ylabel() is used to set the label for y-axis.
- Syntax: plt_obj.ylabel('label')

5.3.5 title() function

- title() is used to set the title.
- Syntax: plt_obj.title('label')

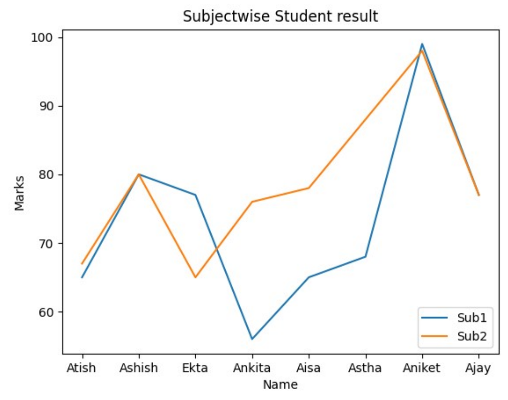
5.3.6 columns function

- Columns displays the list of fields available in given dataframe.
- Syntax: dataframe_object.columns

5.3.7 legend() function

- It is an area describing element of graph.
- It is required when multiple graphs are displayed on one Figure.
- It helps in differentiating multiple graphs.
- Syntax: plt_obj.legend(optional_parameters)
- Parameters:
  - loc='position' : To set the legend position, it is used.
    (Values: 'upper left' (default), 'upper right', 'lower left', 'lower right')
  - ['label1','label2',...]: To set the label in legend.
- Example:

import pandas as pd import matplotlib.pyplot as plt df=pd.read_excel('chart.xlsx') r=df.Rno n=df.Name s1=df.Sub1 s2=df.Sub2 **plt.plot(n,s1)** **plt.plot(n,s2)** plt.title('Subjectwise Student result') plt.xlabel('Name') plt.ylabel('Marks') **plt.legend(['Sub1','Sub2'],loc='lower right')** plt.show()	import pandas as pd import matplotlib.pyplot as plt df=pd.read_excel('chart.xlsx') r=df.Rno n=df.Name s1=df.Sub1 s2=df.Sub2 **plt.plot(n,s1,label='Sub1')** **plt.plot(n,s2, label='Sub2')** plt.title('Subjectwise Student result') plt.xlabel('Name') plt.ylabel('Marks') **plt.legend()** plt.show()



5.3.8 range() function

- It is used to create list based on given parameters.
- Syntax:

  range(start,stop,skip)
- Example:

```
a=range(10)
for i in a:
print(i, end=" ")
b=range(-10,-1)
print('\n')
for i in b:
print(i,end=' ')
print('\n')
c=range(1,10,2)
for i in c:
print(i,end= " ")
```

Output:

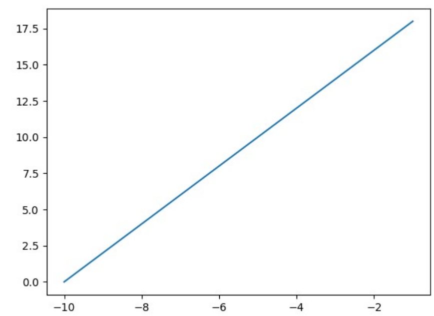
0 1 2 3 4 5 6 7 8 9

-10 -9 -8 -7 -6 -5 -4 -3 -2

1 3 5 7 9

- Example:

```
import matplotlib.pyplot as plt
x=range(-10,0)
y=range(0,20,2)
plt.plot(x,y)
plt.show()
```



5.3.9 len() function

- It returns numbers of Values (length) of given object.
- Syntax: len(object)
- Example:

```
a=[4,5,6,7,8,10]
print(len(a))
b=range(10)
print(len(b))
c=range(1,10,2)
print(len(c))
```

Output:

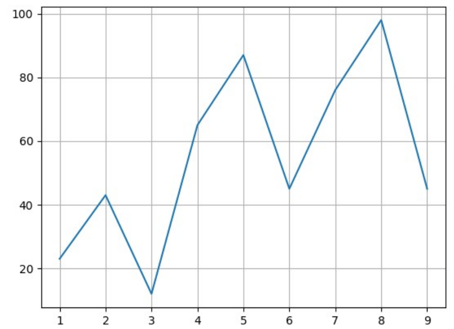
```
6
10
5
```

- Example plot

5.3.10 Grid() function

- It helps to add gridline to the plot.
- Syntax: plt_obj.grid()
- Example:

```
import matplotlib.pyplot as plt
x=range(1,10)
y=[23,43,12,65,87,45,76,98,45]
plt.plot(x,y)
plt.grid()
plt.show()
```

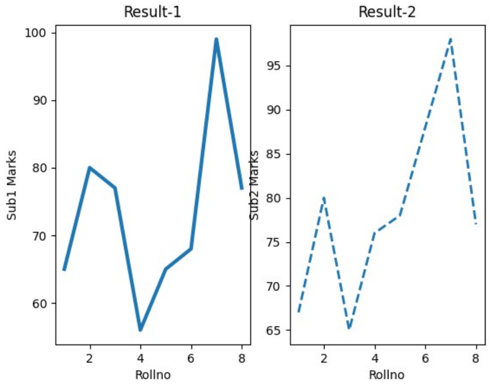


5.3.11 subplot() function:

- Subplot provides an object-oriented approach in data visualization.
- It helps to plot multiple graphs on one figure.
- Syntax: subplot(no_rows, no_cols, sequence_no)
- Means subplot(2,1,1) divides the figure into 2 rows and 1 column. Plot the respective graph in first part.
- Example:

```
import pandas as pd
import matplotlib.pyplot as plt
df=pd.read_excel('chart.xlsx')
plt.subplot(1,2,1)
plt.plot(df['Rno'],df['Sub1'],linewidth=3)
```

```
plt.xlabel('Rollno')
plt.ylabel('Sub1 Marks')
plt.title('Result-1')
plt.subplot(1,2,2)
plt.plot(df['Rno'],df['Sub2'],linewidth=2,linestyle='--')
plt.xlabel('Rollno')
plt.ylabel('Sub2 Marks')
plt.title('Result-2')
plt.show()
```



5.4 Different types of plot graphs:

5.4.1 Line Chart

- Plot() function plot data in form of Line.
- By applying different parameters of plot(), different line style, color, width, market etc. can be changed of Line.

5.4.2 scatter plot

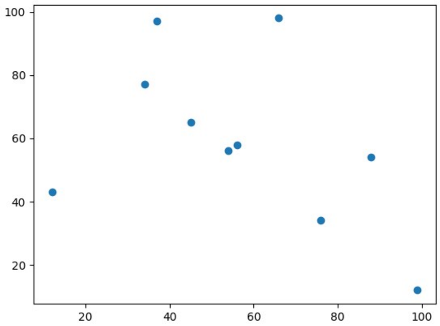
- Scatter plot is used to check the relationship between two set of values.
- Scatter() method of pyplot module is used to draw scatter plot.
- Syntax:

  Matplotlib.pyplot.scatter(x_values,y_values,optional_parameters)
- Optional Parameters:
  - marker='symbol': To change symbol of each point.
  - c='color': To display points with specific color.
  - edgecolor='color': To display points with edge color.
  - s=no : Size of marker
  - linewidths=no : To set line width of border
  - alpha= no : No should be in between 0 to 1. (Here 0 means Transparent and 1 means Opaque). To display the color with some transparency.

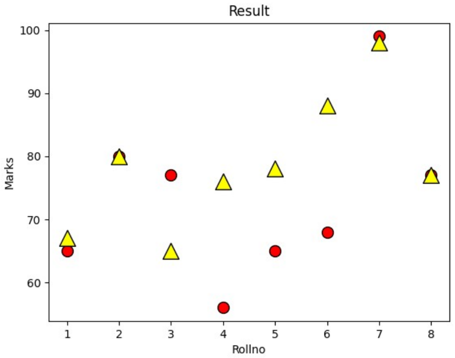
- Example:

```
#Single scatter graph on one axes
import matplotlib.pyplot as plt
x=[12,54,34,66,76,45,99,88,56,37]
```

```
y=[43,56,77,98,34,65,12,54,58,97]
plt.scatter(x,y)
plt.show()
```



```
#Multiple scatter plot on one axes
import pandas as pd
import matplotlib.pyplot as plt
df=pd.read_excel('chart.xlsx')
x=df.Rno
y1=df.Sub1
y2=df.Sub2
plt.scatter(x,y1,marker='o',c='Red',s=100,edgecolor='Black')
plt.scatter(x,y2,marker='^',c='Yellow',s=200,edgecolor='Black')
plt.xlabel('Rollno')
plt.ylabel('Marks')
plt.title('Result')
plt.show()
```



5.4.3 Histogram Chart

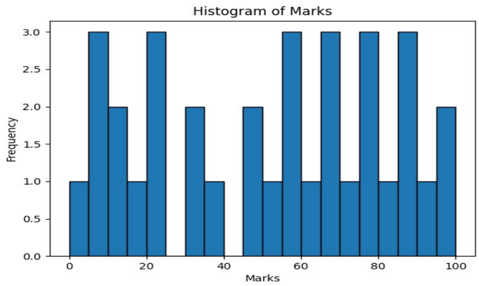
- Histogram is a graphical representation which organize group of data points into specific range.
- It represents the data in form of specific interval.
- It shows interval verses frequency graph.
- hist() method of pyplot module is used to show histogram.
- Syntax:

  matplotlib.pyplot.hist(values, optional_parameters)

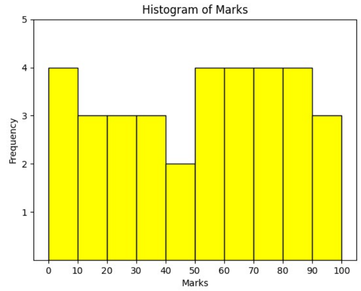
- Optional Parameters:
    - edgecolor='color': To set edge color
    - bins=range : To set the interval manually
    - color='color': To set color

- Example:

```
#Simple histogram program
import pandas as pd
import matplotlib.pyplot as plt
marks=[5,8,12,54,34,66,76,45,99,88,56,37,0,76,88,90,95,58,69,34,23,65,80,60,46,24,
71,16,8,88,55,77,23,12]
plt.hist(marks,edgecolor='Black',bins=20,range=(0,100))
plt.xlabel('Marks')
plt.ylabel('Frequency')
plt.title('Histogram of Marks')
plt.show()
```




```
# Histogram with specific range
import pandas as pd
import matplotlib.pyplot as plt
marks=[5,8,12,54,34,66,76,45,99,88,56,37,0,76,88,90,95,58,69,34,23,65,80,60,46,24,
71,16,8,88,55,77,23,12]
seq=[0,10,20,30,40,50,60,70,80,90,100]
y=[1,2,3,4,5]
plt.hist(marks,edgecolor='Black',bins=seq,color='Yellow')
plt.xlabel('Marks')
plt.ylabel('Frequency')
plt.title('Histogram of Marks')
plt.yticks(y)
plt.xticks(seq)     #To set the values of SEQ on X axis.
plt.show()
```



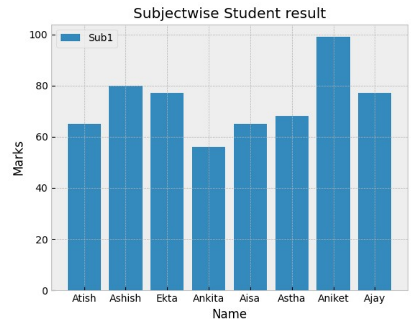
5.4.4 Bar Chart

- Bar chart is used to display graph in form of Bars.
- bar() method is provided by pyplot which helps to plot bar graph.
- Syntax:

  matplotlib.pyplot.bar(x_val,y_val,optional_parameters)

- Optional parameters:
  - Label='text': To set label'
  - color='color': To change color
  - .....
- Example:

```
import pandas as pd
import matplotlib.pyplot as plt
df=pd.read_excel('chart.xlsx')
plt.style.use('bmh')
n=df.Name
s1=df.Sub1
plt.bar(n,s1,label='Sub1')
plt.title('Subjectwise Student result')
plt.xlabel('Name')
plt.ylabel('Marks')
plt.legend()
plt.show()
```



5.5 style() method: (Not in Syllabus)

- pyplot allows multiple styles which can be set to the graph directly.
- To check inbuilt available styles write following:

```
print(plt.style.available)
```

Output:

['Solarize_Light2', '_classic_test_patch', 'bmh', 'classic', 'dark_background', 'fast', 'fivethirtyeight', 'ggplot', 'grayscale', 'seaborn', 'seaborn-bright', 'seaborn-colorblind', 'seaborn-dark', 'seaborn-dark-palette', 'seaborn-darkgrid', 'seaborn-deep', 'seaborn-muted', 'seaborn-notebook', 'seaborn-paper', 'seaborn-pastel', 'seaborn-poster', 'seaborn-talk', 'seaborn-ticks', 'seaborn-white', 'seaborn-whitegrid', 'tableau-colorblind10']

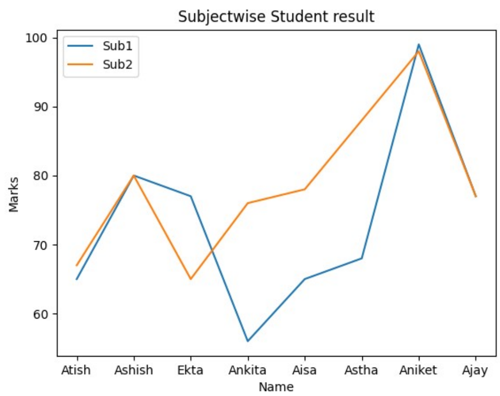
- It is possible to apply using style.use() function.

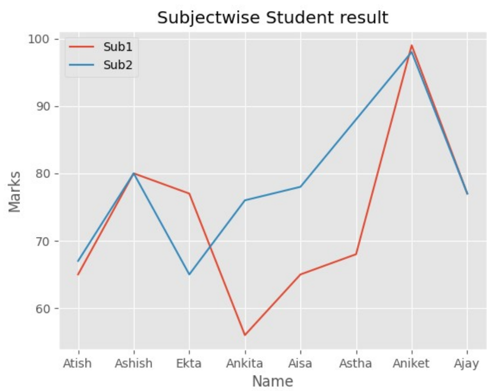
- Syntax:

  matplotlib.pyplot.style.use('Style_name')

- Example: (without Style and With style graph)

```
import pandas as pd
import matplotlib.pyplot as plt
df=pd.read_excel('chart.xlsx')
plt.style.use('ggplot')
r=df.Rno
n=df.Name
s1=df.Sub1
s2=df.Sub2
plt.plot(n,s1,label='Sub1')
plt.plot(n,s2, label='Sub2')
plt.title('Subjectwise Student result')
plt.xlabel('Name')
plt.ylabel('Marks')
plt.legend()
plt.show()
```





5.6 Saving chart: (Not in Syllabus)

- To save the graph as .png file, savefig() method is used.
- Syntax:

  Matplotlib.pyplot.savefig('filename')

- Example:

  Plt.savefig('test.png')

5.7 Plotting using DataFrame_object.plot()

- Direct using dataframeplot(), graph can be plotted.
- Dataframe.plot() function is used to plot the graph.
- Syntax:

  .plot(x=values, y=values,kind='chart_type', optional_parameters)

- Optional Parameters:
  - Kind='chart_type': To specify which graph you want to plot.
  - Marker

  - Linestyle
  - Linewidth
  - Color etc…….

- Example:

1. Plot line chart

```
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_excel ('chart.xlsx')
df.plot(x ='Name', y=['Sub1','Sub2'], kind = 'line', color='r', linewidth=2, marker='o', linestyle='--')
plt.show()
```

2. Plot bar chart

```
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_excel ('chart.xlsx')
df.plot(x ='Name', y=['Sub1','Sub2'], kind = 'bar',color='pink',linewidth=2,linestyle='--',edgecolor='black')
plt.show()
```

3. plot scatter chart

```
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_excel ('chart.xlsx')
df.plot(x ='Rno', y='Sub1', kind = 'scatter',marker='o')
plt.show()
```